

House Price Prediction

Machine Learning Project - Findings Summary
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01 - PROJECT OVERVIEW :

This project develops a supervised machine learning pipeline to predict residential house prices using property attributes. The dataset, sourced from Kaggle, contains **545 records** with **13 features** spanning physical dimensions (area, bedrooms, bathrooms), structural attributes (stories, basement, hot water heating), and amenity indicators (main road access, air conditioning, parking, preferred area, furnishing status). The target variable is **price** (in Indian Rupees). The end goal is a deployable baseline model for real estate pricing decisions.

02 - DATASET AT A GLANCE :

No missing values were found. Duplicate rows were removed. All binary categorical columns (yes/no) and furnishing status (furnished / semi-furnished / unfurnished) were one-hot encoded, expanding the feature set from 12 to 16 predictors for model training.

03 - MODEL RESULTS (80 / 20 Train-Test Split):

Model	RMSE	MAE	R ² Score	Verdict
Linear Regression	13,24,507	9,70,043	0.6529	Best
Random Forest (100 trees)	14,00,566	10,21,546	0.6119	Runner-up

observation: Linear Regression outperforms Random Forest on this 545-row dataset, explaining **65.3%** of price variance ($R^2 = 0.6529$). Ensemble methods like Random Forest typically require larger training sets to surpass a well-specified linear model - a textbook finding confirmed here. The MAE of 9,70,043 means predictions are on

average within ~9.7 lakh of the actual price, which is acceptable for a baseline estimation tool.

04 - KEY FINDINGS & INSIGHTS :

- **Area is the Dominant Price Driver**

With a Random Forest feature importance score of 0.468, property area accounts for nearly half of all predictive power. A 1,000 sq ft increase in area has the largest single positive impact on price, making it the primary metric for valuation.

- **Bathrooms & Amenities Outweigh Bedrooms**

Bathrooms (importance 0.151) rank second - significantly ahead of bedrooms. Air conditioning (0.063) and parking (0.057) rank higher than bedroom count, indicating buyers prioritize comfort and utility over raw room quantity.

- **Location Premium is Real**

Properties on the main road and in preferred areas command measurable price premiums. These binary flags carry consistent positive coefficients in the Linear Regression model, confirming that location access boosts market value independently of physical size.

- **Furnishing Status Adds Layered Value**

Furnished homes are priced higher than semi-furnished, which in turn exceeds Unfurnished. The one-hot encoding separates these tiers cleanly, and both dummy variables carry positive weight in the regression, validating this Hierarchy.

05 – VISUALISATIONS:

House Price Distribution Analysis



Fig 1 - Price Distribution: Right-skewed histogram + box plot revealing outliers above 10M

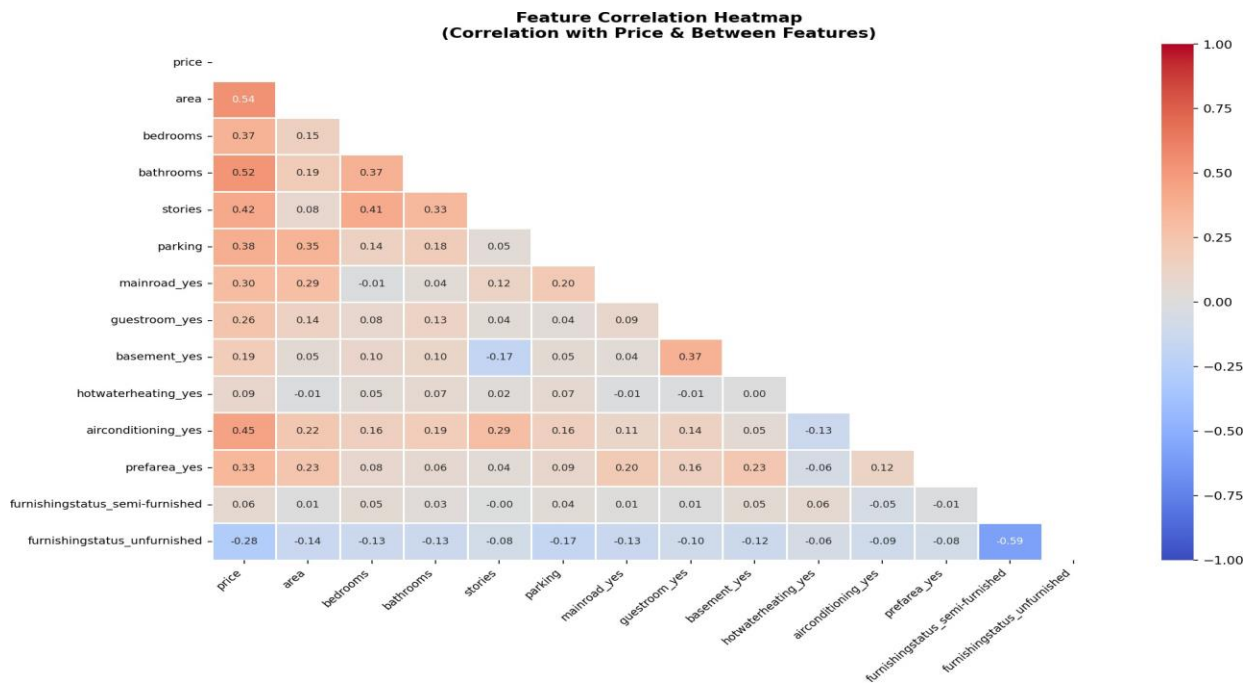


Fig 2 - Correlation Heatmap: Area, bathrooms, and stories show strongest positive correlation with price

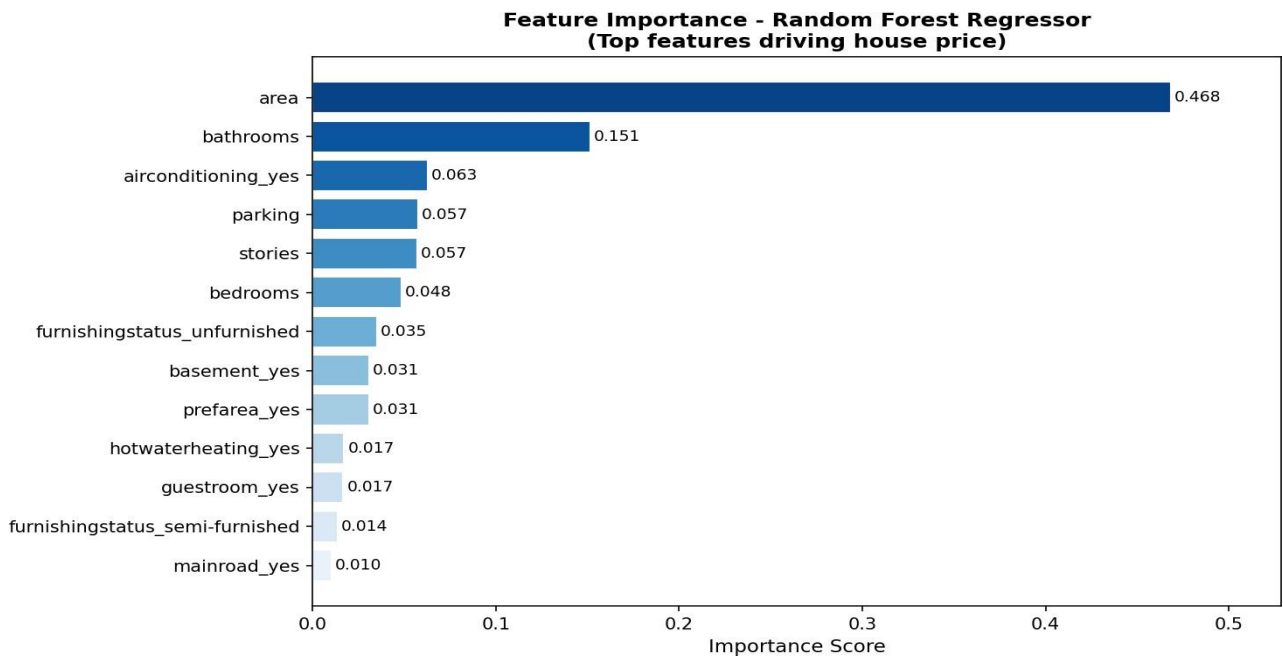


Fig 3 - Feature Importance (RF): Area dominates; air conditioning ranks above bedroom count

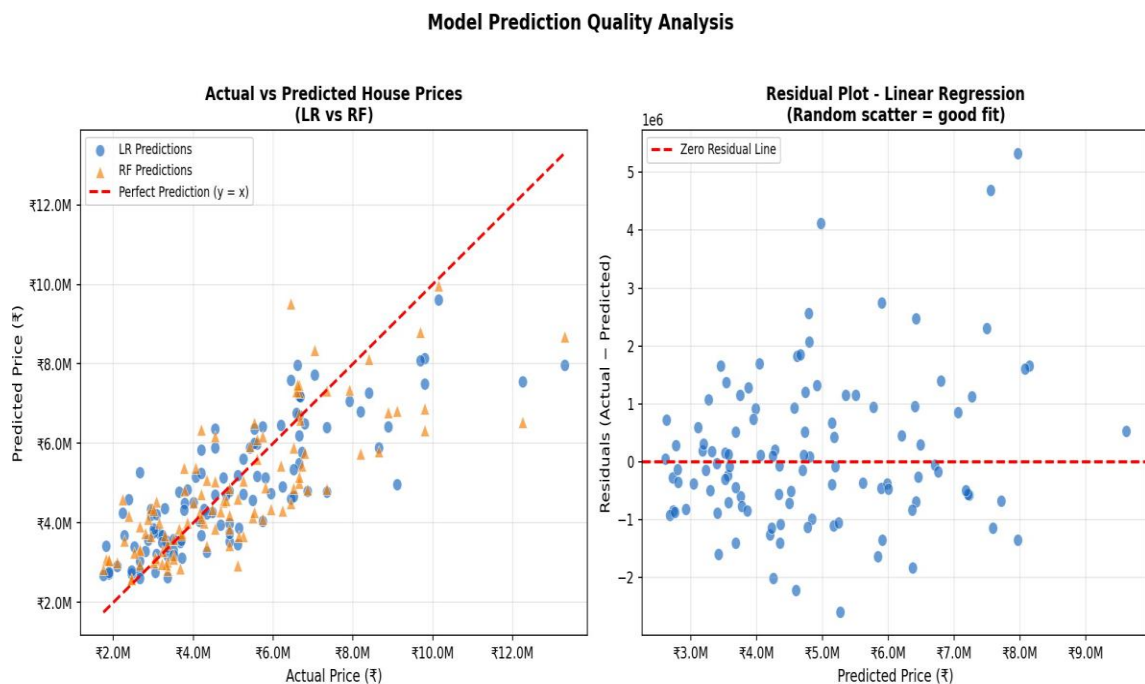


Fig 4 - Actual vs Predicted: Both models track the diagonal; residuals spread randomly (good fit)

Conclusion: A complete ML pipeline was built covering data ingestion, cleaning, one-hot encoding, modelling, evaluation, and visualization. Linear Regression ($R^2 =$

0.6529) emerged as the best-performing model, outperforming Random Forest on this 545-record dataset. **Area, bathrooms, air conditioning, and parking** are the top four features driving house prices. The model is suitable as a production baseline for real estate price estimation and can be improved with more data and feature engineering (log-transform of price, interaction terms, or gradient boosting).